Enhancing Code Retrieval Accuracy in Developer-Oriented RAG Systems using Semantic Chunking

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Introduction

What is RAG?

Retrieval-Augmented Generation (RAG) augments LLMs with an external knowledge retrieval step, enabling domain-specific responses without costly retraining. It has proven highly effective in text-based NLP tasks including QA, fact-checking, and summarization.

The Code Retrieval Gap

Applying RAG to source code is fundamentally challenging. Standard text chunking (fixed character-size) is content-agnostic and breaks code at arbitrary points — cutting through function bodies, class definitions, and docstrings — producing fragments that are syntactically incomplete and semantically misleading.

This Research

We compare two RAG chunking pipelines over the Flask Python codebase (83 source files): Pipeline A uses fixed-size character splitting while Pipeline B uses AST-aware, Python-syntax-aligned semantic chunking. Both use the same embedding model (all-MiniLM-L6-v2) and ChromaDB vector store.

Problem Statement

Core Problem

Standard RAG pipelines split source code by fixed character count (chunk_size=1000), ignoring Python syntax. The offset k in c_i = S[k : k + chunk_size] is determined purely by character count — not by semantic or syntactic boundaries.

Resulting Failure Modes

- **Context Fragmentation:** Chunks start mid-function or mid-class, losing the function signature and docstring that LLMs need for understanding.
- **Keyword Blindness:** When two functions are merged in one chunk, shorter keyword-dense functions outrank the intended target, diluting embedding precision.
- **Wrong File Retrieval:** Test files containing related method names can rank above the source implementation, returning irrelevant scaffolding code.

Research Question

Does substituting fixed-size chunking with AST-aware, Python-syntax-aligned semantic splitting improve code retrieval quality in a developer-focused RAG system?

Objective

This study aims to:

- Perform a **controlled comparison** of fixed-size vs. AST-aware chunking strategies over a real-world Python codebase (Flask web framework, 83 source files).
- Identify and **characterize failure modes** (context fragmentation, keyword blindness) in baseline character-based RAG retrieval.
- Provide **empirical evidence** that semantic chunking produces syntactically complete, self-contained code units that improve retrieval precision.
- Demonstrate an **open-source, reproducible experimental framework** using LangChain, ChromaDB, and HuggingFace — no proprietary services required.
- Evaluate retrieval across **20 developer-focused natural language queries** covering method behaviour, class structure, lifecycle, configuration, and CLI patterns.
- Show that AST-aware chunking is a **zero-cost, drop-in upgrade** over any existing LangChain-based RAG pipeline for code.

Literature Review

RAG Foundations

- **Lewis et al. [2020]:** Introduced RAG framework fusing dense passage retrieval with seq2seq generation; no code support.
- **Asai et al. [Self-RAG, 2024]:** LLM decides when to retrieve and whether passages are useful; reduces irrelevant context in output.

Code-Specific RAG & Chunking

- **Wang et al. [CodeRAG-Bench, 2025]:** Found chunk sizes 200–800 tokens optimal; no AST awareness evaluated.
- **Ding et al. [CrossCodeEval, 2023]:** Function-level granularity outperforms file-level and line-level for cross-file code completion; no comparison study.
- **Zhang et al. [RepoCoder, 2023]:** Iterative function-level retrieval for repository code using BM25; no controlled comparison.
- **Zhou et al. [DocPrompting, 2023]:** Document-level retrieval for library docs; no function isolation in chunks.

Research Gap Addressed

No prior study simultaneously applies AST-aware semantic chunking to a Python library corpus stored in ChromaDB with a controlled side-by-side comparison — this paper fills that gap.

Methodology

System Architecture (4 Stages)

Stage 1: Document Loader

LangChain DirectoryLoader + TextLoader ingests all 83 Flask .py source files, producing one Document object per file.

Stage 2: Chunking Module (differs per pipeline)

The single variable component — held constant everywhere else to isolate chunking effects.

Stage 3: Embedding Module

all-MiniLM-L6-v2 (22.7M parameters, 384-dim dense vectors). Identical for both pipelines.

Stage 4: ChromaDB Vector Store

Two separate persistent collections. Retrieval via cosine similarity, k=1 (strict single-best-match comparison).

Dataset

Flask web framework (v3.x) — 83 .py files including app.py, blueprints.py, cli.py, config.py, ctx.py, wrappers.py, and test files. Diverse constructs: class hierarchies, decorators, context managers, type annotations.

Hyperparameters

chunk_size: 1000 characters

chunk_overlap: 200 characters

Retrieval k: 1 (strict comparison)

Embedding dim: 384 (fixed by MiniLM)

Similarity: Cosine (ChromaDB default)

Queries: 20 developer-focused NL questions

Scoring Rubric

- **0 — Fail:** Wrong function or missing signature
- **1 — Partial:** Correct file, mid-body fragment
- **2 — Correct:** Full signature + docstring at chunk start

Methodology: Pipeline A vs Pipeline B

Pipeline A — Baseline

Fixed-Size Character Chunking

Splitter: RecursiveCharacterTextSplitter

Chunks produced: 804

Split logic: $c_i = S[k : k + chunk_size]$ where k is an arbitrary character offset. Tries paragraph/newline/whitespace breaks; falls back to character cut.

Code awareness: None — ignores Python syntax entirely.

Drawback: Frequently bisects function bodies and class definitions, merging logically distinct code units to fill character quotas.

Result: Boundary alignment is **incidental** — correct only when a function happens to coincide with a character-count boundary.

Pipeline B — Proposed

AST-Aware Semantic Chunking

Splitter: RecursiveCharacterTextSplitter.from_language(Language.PYTHON)

Chunks produced: 861 (+7.1% vs. baseline)

Split hierarchy: (1) class definitions → (2) function defs → (3) blank lines → (4) single newlines → (5) character level.

Formal: $c_j = AST_node(f), f \in \{class_def, func_def, method_def\}$

Key advantage: Each function node encapsulates its complete body, decorators, signature, and docstring — exactly what an LLM needs to understand context.

Result: Boundary alignment is **consistent by construction** — every chunk starts at a class or function definition.

Results & Discussion

Chunk Statistics Comparison

Source files (both): **83 .py files**
Pipeline A chunks: **804**
Pipeline B chunks: **861 (+7.1%)**
Critical query failures A: **4 documented**
Critical query failures B: **0**
Boundary retrieval A: **Incidental**
Boundary retrieval B: **Consistent**

Aggregate Score (20 Queries)

28/40

Pipeline A (70%)

40/40

Pipeline B (100%)

Key Discussion Points

- The 57-chunk increase (+7.1%) reflects the AST splitter's refusal to merge semantically distinct code units just to fill a 1,000-character limit. Each extra chunk represents a code unit that would have been improperly merged in the baseline.
- Smaller, semantically consistent chunks improve embedding precision — each vector represents an increasingly specific semantic unit. Aligns with Wang et al. [3] who found 200–800 token chunks enable better retrieval.
- The only cost is a negligible storage/indexing overhead for 57 extra chunks — no additional dependencies beyond standard LangChain, making it a zero-cost drop-in upgrade.

Critical Failure Modes in Pipeline A

Four cases where Pipeline A returned qualitatively incorrect results. Pipeline B resolved all four by design:

F1 — Keyword Blindness: copy_current_request_context

Pipeline A: Retrieved def copy(self) — WRONG function. Fixed-size chunk merged both functions; shorter copy(self) was more keyword-dense, diluting the embedding for the intended target.

Pipeline B: Each function isolated in its own chunk — exact copy_current_request_context(f:F) retrieved.

F2 — Context Fragmentation: _AppCtxGlobals class

Pipeline A: Mid-docstring fragment; class definition missing entirely — useless for an LLM needing class context.

Pipeline B: class _AppCtxGlobals: retrieved with full docstring from the class definition boundary.

F3 — Context Fragmentation: has_request_context()

Pipeline A: Bottom half of docstring only; no signature — the LLM cannot determine what function this belongs to.

Pipeline B: def has_request_context() -> bool: with full signature and usage example code block.

F4 — Context Fragmentation: ConfigAttribute descriptor

Pipeline A: rv = obj.config[...] — mid-method fragment. No class definition or __init__ visible.

Pipeline B: class ConfigAttribute(t.Generic[T]): with complete __init__ and all property descriptors.

Per-Query Retrieval Scores

Score: 0=Fail 1=Partial 2=Correct

Query Target	A	B
get_send_file_max_age	2	2
open_resource method	1	2
copy_current_request_context	0	2
_AppCtxGlobals class	0	2
has_request_context()	0	2
ConfigAttribute descriptor	0	2
load_dotenv prioritization	2	2
AppContext.pop() lifecycle	2	2
create_url_adapter	2	2
from_prefixed_env method	2	2

(continued)

Query Target	A	B
send_static_file Flask vs BP	1	1
on_json_loading_failed debug	2	2
Response max_cookie_size	0	2
find_best_app process	1	2
ScriptInfo.load_app	0	1
with_appcontext decorator	2	2
load_dotenv file priority	2	2
Config.from_prefixed_env	2	2
ConfigAttribute descriptor fetch	0	2
from_pyfile vs from_object	2	2

TOTAL: A = 28/40 (70%) B = 40/40 (100%)

Conclusion & Future Work

Key Findings

- **Finding 1 — Chunk Quality:** Pipeline B produced 861 syntactically meaningful chunks vs. 804 arbitrary fragments in Pipeline A. Every semantic chunk contained a self-contained, complete code unit.
- **Finding 2 — Failure Elimination:** All four critical failure modes in Pipeline A (context fragmentation F2, F3, F4 and keyword blindness F1) were entirely eliminated by the structural guarantees of AST-aware chunking.
- **Finding 3 — Retrieval Score:** Pipeline B achieved a **perfect 40/40 (100%)** aggregate score vs. 28/40 (70%) for the baseline — a 30-point improvement.
- **Core Conclusion:** Code structure must be a first-class concern in RAG preprocessing. Traditional text-based chunking produces unacceptable retrieval quality for developer-oriented use cases.

Future Work

- **End-to-End LLM Generation:** Add a generation stage for quantitative pass@k evaluation alongside the qualitative analysis.
- **Code-Specific Re-ranking:** Apply models such as voyage-code-2 or Jina-v2-code on top of AST-aware chunks to further boost retrieval precision.
- **Broader Codebase Validation:** Evaluate on Django, NumPy, and other large Python projects to confirm generalisability beyond Flask.
- **Hybrid Retrieval:** Combine BM25 lexical search with dense AST-aware semantic search to mitigate residual ambiguity in embedding space.
- **Multi-Language Support:** Extend to TypeScript, Java, and C++ to broaden applicability across the full spectrum of developer tooling.

Thank You

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